Daily Drilling Report
SPD19B-03
4/7/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
13.0	16"	34.00	72.83	31.70	100.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
18 5/8" Shoe, 870.00mKB	199,111.37	7.80	3,121,429.60		10.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
875.00	875.00	0.00			

Operations at Report Time
Work on stuck pipe.

Operations Summary
N/U 21 1/4" BOP. BOP test. RIH 16" rotary BHA, tag cement @ 771m. D/O cement inside casing and tag float collar @ 845m.

Operations next 24hrs
D/O 18 5/8" shoe track. Work on stuck pipe and make string free. Cont' D/O shoe track and 3m new formation, perform FIT. POOH to surface.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari	Night Drilling Sup.
Behnam Jahangiri	Drilling Engineer
Ali Sahebnouri	Geologist
M. H. Hemmati	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	4/9/2014
BOP Test		
Chemical Spill Drill	3/29/2014	5/29/2014
Fire Drill	3/9/2014	4/9/2014
H2S Drill	4/4/2014	5/5/2014
Heli Crash Drill	3/22/2014	5/22/2014
Medical Drill	4/4/2014	6/4/2014
MOB Drill	2/28/2014	4/30/2014
Pit Drill		
Trip Drill		

STOP Cards

Number of Reports: 30

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	01:00	1.00	16"	C	05	P	N/U 21 1/4" BOP Kill coflex hose.
01:00	01:30	0.50	16"	C	05	P	Remove scaffold from kill coflex and connect BOP hydraulic hoses.
01:30	03:00	1.50	16"	C	05	P	M/U Riser (20 3/4" 3K drive lock bottom x 20 3/4" 3K flange top) on CHH.
03:00	04:00	1.00	16"	C	05	P	Build scaffold around well B-03.
04:00	06:30	2.50	16"	C	05	P	N/U DSA (20 3/4" 3K Studded bottom x 21 1/4" 2K Studded top) on riser. N/U 21 1/4" BOP on DSA. Tighten the bolts.
06:30	09:30	3.00	16"	C	05	P	N/U 7m Mandrell + Overshot + Diverter packer assembly. >>> Final 21 1/4" BOP Schematic (f/ top to bottom): * Diverter assembly , 3.32m * Overshot , 4.7m * 7m PPL Mandrel (21 1/4" flange btm) * 21 1/4" BOP stack, 3K blind + 2K annular, (21 1/4" flange top and btm) * PPL DSA (20 3/4" 3K studded btm x 21 1/4" 2K studded top) , 0.3m * PPL Riser (20 3/4" 3K drive lock btm x 20 3/4" 3K flange top) , 3.24m * Wellhead CHH (20 3/4" Hub top) > Use RX-73 gasket between CHH and riser. > Use RX-74 gasket between riser and DSA. > Use RX-73 gasket between DSA and BOP. > Use RX-73 gasket between BOP and Mandrel. >> Use BX-154 gasket between BOP block adaptors and coflex hoses. >> Use BX-155 gasket between BOP manual gate valve and block adaptor. ** CHH landing shoulder - RKB = 19.45m
09:30	13:30	4.00	16"	C	12	P	M/U BOP test assembly consists of: Test Plug + XOS + 2 jts 5 7/8" DP + SES + FOSV + 15 ft pup joint, RIH to 19.45m. Line up to cement unit. Test surface lines to 2000 psi, OK. Pressure test Annular BOP + Kill and choke manual valve + Kill and choke HCR + Kill and choke hoses to 500 / 1500 psi , OK. Back off test plug inside CHH. Pressure test Blind ram to 500 / 1500 psi , OK. Retreive test plug, POOH and L/D test assy.
13:30	14:00	0.50	16"	C	01	P	M/U wear bushing RT and install wear bushing inside CHH.

Daily Drilling Report
SPD19B-03
4/7/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
14:00	17:30	3.50	16"	D	04	P	P/U and M/U shoe track rotary BHA consists of: 16" Mill Tooth bit (CR3GJMRS, IADC: 135, SN: 255758, Nozzles: 3 x 22 + 1 x 16) + Bit sub with float valve + 15 3/4" Stabilizer + 1 jt 9 1/2" DC + XOS + 2 jts 8 1/4" DC + XOS + 6 STD 5 7/8" HWDP, RIH to 201m. Cont' RIH BHA on 5 7/8" DP and tag cement @ 771m.
17:30	00:00	6.50	16"	D	03	P	Drill out cement and cement plugs inside 18 5/8" casing from 771m w/ 800 - 1000 gpm/ 5 - 25 kips/ 780 - 900 psi/ 45 rpm/ 2 - 12 kft-lbf. Tag float collar @ 845m. ** Pump 20 bbl Hi-Vis pill @ tool joints. ** Pump 40 bbl Hi-Vis pill when stand down, meanwhile B/R and R/D the stand.
00:00	00:00						Apr 08 , 2014 00:00 - 03:00 Drill out float collar @ 845m, cement inside shoe track to 867.5m w/ 900 - 1000 gpm/ 700 - 800 psi/ 5 - 30 kips/ 45 rpm/ 3 - 15 kft-lbf. ** Pump 40 bbl Hi-Vis pill after drill out float collar. ** Pump 40 bbl Hi-Vis pill while drilling @ 857m. ** Pump 40 bbl Hi-Vis pill when stand down @ 867.5m. ** Pull string inside float collar and check with & without rotation, Ok. 03:00 - 06:00 After connection @ 867.5 , try to resume drilling, suddenly pressure shoot up. String got packed off and stuck. Work on stuck by moving down while keep torque inside string. no success.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
875.00	Sea Water	8.60				
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.40	7.	340	2.0	1,013.0	24.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Osam Condor	Supply Boat	4/7/2014 06:30	4/7/2014 16:45	Receive B/L from rig. Departed to Kish.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Barite Bulk	3.0	103.0	96.0
Lime	2.0	338.0	141.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Barite BB	0.0	0.0	0.0
Cement G	0.0	0.0	39.0
Bentonite Bulk	0.0	0.0	25.0
Bentonite BB	0.0	0.0	23.0
Barite Bulk	3.0	25.0	88.0
Drill Water	0.0	50.0	216.0
Diesel Fuel	11.0	0.0	115.0
Potable Water	20.0	20.0	110.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
4	16	Varel	CR3GJMRS	255758	Mill Tooth	135	22/22/22/16
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
1.31	0.00						-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
6 STD 5 7/8" HWDP	5 7/8	4.250	167.40	201.19
XOS	7	2.810	0.79	33.79
2 jts 8 1/4" DC	8 1/4	2.810	18.90	33.00
XOS	9 1/4	3.000	1.12	14.10
1 jt 9 1/2" DC	9 1/2	3.000	9.45	12.98
15 3/4" Stabilizer	15 3/4	3.500	2.17	3.53

Daily Drilling Report
SPD19B-03
4/7/2014

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
Bit sub with float valve inside	9 1/8		0.95	1.36
16" Mill Tooth Bit	16		0.41	0.41

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)

Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Asmari		374.50	372.44
Jahrum		493.00	487.29
Ilam		847.00	

General Remarks

Comment

>>> Requirements: (200 MT Cement)

>>> Sent:

** Osam Condor to Kish: (1 x 18 5/8" top cement plug) + (1 x 18 5/8" bottom cement plug) + (5 x 18 5/8" casing coupling) + (2 no 18 5/8" float shoe) + (2 no 18 5/8" float collar) + (3 no 24" bit) + (1 no CHH) + (38 no bow centralizer) + (18 5/8" Casing handling tools) + (24" section wellhead tools).

>>> Received:

** Miami from kish: 25 MT Barite
** Osam Condor from Kish: 50 MT DW